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Odd End Semester Examination December- 2025

Name of the Program: **B. Tech Biotechnology**
 Name of the Course: **MICROBIOLOGY**
 Time: 3 hrs

Semester: III
 Course Code: **TBT 303**
 Maximum Marks: 100

Note: -

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty
- (iv) Each question carries 10 marks

Q1	(20 marks)	
(a)	Describe the structural and functional significance of the murein layer in maintaining bacterial cell wall integrity.	CO1
(b)	Explain the purpose and effect of the decolorizing agent used during Gram's staining in differentiating bacterial species.	
(c)	Distinguish between the mechanisms of generalized and specialized transduction in bacteria, citing appropriate examples.	
Q2	(20 marks)	
(a)	Compare the principles and applications of numerical taxonomy with molecular-based classification techniques in Microbiology.	CO2
(b)	Elaborate on Robert Koch's contributions to the establishment of the germ theory of disease and its scientific importance.	
(c)	Identify one prominent scientist from the Golden Age of Microbiology and briefly describe their key contribution to the field.	
Q3	(20 marks)	
(a)	Explain the concept and importance of functional classification of culture media used in Microbiology, providing relevant examples.	CO3
(b)	Discuss the physiological and environmental factors responsible for the slowdown of bacterial growth during the stationary phase.	
(c)	A 1 mL water sample suspected of containing coliforms is provided. Describe the experimental steps to detect and confirm the presence of coliform bacteria in the laboratory.	
Q4	(20 marks)	
(a)	Illustrate and describe the various phases of a bacterial growth curve, mentioning the characteristic features of each phase.	CO4
(b)	Explain the procedure for propagation of animal viruses using cell lines and discuss its significance in Virology.	
(c)	Design an experiment to demonstrate the influence of temperature on bacterial growth and present the expected graphical outcome.	
Q5	(20 marks)	
(a)	Discuss the types of sterilization procedures employed to maintain aseptic conditions inside a laminar airflow cabinet.	CO5
(b)	Explain different chemical methods of sterilization and mention their specific uses in a Microbiology laboratory.	
(c)	Define pure and mixed cultures. Describe the laboratory techniques used to obtain a pure culture from a mixed population of microorganisms.	

